Client Check in
Section 1 has the data loading and the column dictionary, section 2 is the EDA, and section 3 has my supervised versus unsupervised decision for both problems. The modelling is Checkpoint 2 work so it isn't in this notebook yet. I ran a first pass already which is saved in working/, because the EDA turned up a problem I wanted to fix before committing to a direction.

The Data

gym_ghosts_master.csv, 50,000 members and 24 columns, from the Kaggle link in the client proposal. It's clean. No missing values, no duplicate ids, no duplicate rows. The only judgement call was the outlier tails and I kept all 50,000 rows, since a 40 km commute and a $101.89 plan are both plausible.

What I Found

Churn is 45.12%, 22,560 out of 50,000, which matches the "nearly half" in the brief.

1) Visit decay - The members who ghost were already showing up half as often in week 1, 1.85 visits versus 3.59. Both groups then decay at the same rate, week 4 is 58% of week 1 for both. This tells us the gap is there from day one and doesn't open up over the month.

2) Feature separation - Only avg_weekly_visits really separates the two groups, with a KS of 0.421. Fee is 0.096 and distance is 0.091, which are shifts and not separations. This tells us there's a ceiling on any model here before I even fit one.

3) Fee and contract type - The raw fee gradient is about half contract mix. Month to month is both the most expensive plan and the least sticky, so churn by fee quintile mostly flattens once you hold contract type constant. Promotion doesn't do the same thing.

The Direction

Both problems are supervised. Problem 1 is explanatory since it asks which of the three options the data supports, so group comparisons, effect sizes and a permutation test. Problem 2 is logistic regression, picked over a tree ensemble because the client needs the fee effect in dollars and not a feature ranking. Reasoning in 3.1 and 3.2.

What's Next
- Compare the three options (churn rates, reach, effect sizes)
- Fit the logistic regression and build the retained revenue curve
- Assumptions log, limitations, executive summary
- Streamlit app